

## Scalability & Future Plan

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|----------------------|-------------------------------------------------------------------|
| <b>Date</b>          | 15 July 2026                                                      |
| <b>Team ID</b>       | SL-2026-02                                                        |
| <b>Project Name</b>  | Smart Lender – Loan Eligibility Prediction Using Machine Learning |
| <b>Maximum Marks</b> | 1 Mark                                                            |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                                           | Impact | Priority to Address |
|------|--------------------------------------------------------------------------------------|--------|---------------------|
| 1    | Limited training dataset reduces prediction accuracy for diverse applicant profiles. | High   | High                |
| 2    | Single machine learning model may not perform optimally for all scenarios.           | Medium | High                |
| 3    | Web application is designed for limited concurrent users and local deployment.       | Medium | Medium              |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                 | Proposed Upgrade / Solution                                                                                                                 |
|------|--------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports a limited number of users.           | Deploy the application on cloud platforms with auto-scaling and load balancing to handle thousands of users simultaneously.                 |
| 2    | Data Storage       | Stores data in a local database or CSV files. | Migrate to scalable cloud databases such as PostgreSQL, MySQL, or MongoDB with backup and recovery mechanisms.                              |
| 3    | Performance        | Predictions are processed sequentially.       | Optimize model inference, implement caching, and use GPU acceleration where applicable for faster predictions.                              |
| 4    | Security           | Basic authentication and validation.          | Implement role-based access control, data encryption, secure APIs, multi-factor authentication, and compliance with data privacy standards. |

**Future Roadmap:**

| Phase   | Planned Feature / Enhancement                                                                           | Target Timeline | Expected Impact                                                        |
|---------|---------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------|
| Phase 1 | Improve model accuracy using advanced algorithms such as XGBoost, Random Forest, and Gradient Boosting. | 2 Months        | Higher prediction accuracy and better decision-making.                 |
| Phase 2 | Develop a responsive web and mobile application for loan eligibility prediction.                        | 4 Months        | Improved accessibility and user experience.                            |
| Phase 3 | Integrate real-time credit score APIs and banking data for enhanced prediction.                         | 6 Months        | More reliable and personalized loan eligibility assessments.           |
| Phase 4 | Deploy the application on cloud infrastructure with continuous model retraining and monitoring.         | 9–12 Months     | Enterprise-level scalability, reliability, and continuous improvement. |

**Expected Long-Term Benefits**

- Support thousands of concurrent users with minimal latency.
- Improve prediction accuracy through continuous machine learning model updates.
- Enhance security using modern authentication and encryption techniques.
- Enable seamless integration with banking systems and financial institutions.
- Provide real-time, data-driven loan eligibility decisions.
- Deliver a scalable, reliable, and intelligent lending solution suitable for financial organizations.